

PHYS 580 Homework 5 (Chap.7)

All questions are worth 10 points, irrespective of complexity or length. Please submit all your solutions on Brightspace in PDF format, together with the source code (if any). Make sure to

- a) discuss the physics and your results, as appropriate and/or directed,
- b) include graphical output to illustrate the results,
- c) include the source codes of your programs (at least the critical parts thereof),
- d) describe briefly what your program does, and how it does it (algorithm), and
- e) state the nature of the numerical approximation you used and demonstrate how you know that the approximation (with the parameters you used) is adequate for the problem you used it for.

Note the errata pages for the textbook: <http://www.physics.purdue.edu/~hisao/book/www/errata.pdf>
which are explained further on <http://www.physics.purdue.edu/~hisao/book/www/errata.html>

1. Problem 7.2 (p.188)

Simulate a random walk in three dimensions allowing the walker to make steps of unit length in random directions; don't restrict the walker to sites on a discrete lattice. Show that the motion is diffusive, that is, $\langle r^2 \rangle \sim t$. Find the value of the proportionality constant.

2. Problem 7.6 (p.194).

Simulate SAWs in three dimensions. Determine the variation of $\langle r^2 \rangle$ with step number and find the value of ν , where this parameter is defined through the relation $\sqrt{\langle r^2 \rangle} \sim At^\nu$.

Compare your results with those in Figure 7.6. You should find that ν decreases with successively higher dimensions. (It is 1 in one dimension and $3/4$ in two dimensions.) Can you explain this trend qualitatively?

Also check whether you can reproduce the analytic result $\nu = 3/4$ for SAW on a 2D grid.

3. Problem 7.12 (p.205).

Calculate the entropy for the cream-in-your-coffee problem, and reproduce the results in Figure 7.16.

If you have trouble with running time, then try first on coarser grids (e.g., 50×50 for the random walk, 4×4 for the entropy, and only 100 particles).

4. Problem 7.15 (p.205)

Perform the random-walk simulation of spreading cream (Figures 7.13 and 7.14), and let one of the walls of the container possess a small hole so that if a cream particle enters the hole, it leaves the container. Calculate the number of particles in the container as a function of time. Show that this number, which is proportional to the partial pressure of the cream particle varies as $\exp(-t/\tau)$, where τ is the effective time constant for the escape.

Hint: Reasonable parameter choices are a 50×50 container lattice and hole 10 units in length along one of the edges.

5. Problem 7.30 (p.228)

Generate a spanning cluster for a two-dimensional square lattice at $p = p_c$ using any of the search methods discussed in connection with Figure 7.29. Estimate the fractal dimensionality of the cluster. You should find a value of d_f , which is slightly smaller than 2 (the expected value is $91/48 \approx 1.90$).